Yimin Gao

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EDUCATION

Ph.D., Electrical Engineering, University of Virginia 2023–2027

M.S., Electrical Engineering, University of Virginia 2021–2022

B.E., Electrical Engineering, Univ. of Cincinnati & Chongqing Univ., Dual Degree Program 2015–2020

SUMMARY

Ph.D. student in computer architecture in AI accelerators and HW/SW co-design for LLM inference. **First-author MICRO 2026 paper**, **ICCD 2026 Best Paper Award**, and **two GF12 tapeouts in progress**. Experience spanning accelerator architecture, RTL design, physical implementation, and system-level performance modeling.

PROJECTS

*FlexPosit: Tunable Fractional Precision for **LLM Inference** Accelerators* **MICRO 2026; GF12**

- **RTL/ASIC Design:** Designed and verified a precision-configurable, bit-serial systolic accelerator in SystemVerilog; validated on FPGA and synthesized in GF12 for Dec. 2026 tapeout.
- **Quantization:** Developed a PyTorch mixed-precision quantization pipeline for LLMs, achieving near-FP16 accuracy at sub-5-bit precision.
- **Architecture Modeling:** Built a cycle-accurate system-level simulator integrating compute, DMA, DRAM, and energy models; demonstrated **1.8× throughput** and **2.0× energy efficiency**.

*FlexNPU: Full-Stack FPGA NPU Platform **(Architecture, Compiler, Runtime)*** **2026 LAD Fellow**

- **NPU Architecture:** Integrated a RISC-V controller, systolic matrix engine, and vector unit for broader ML operator coverage, with on-chip scratchpad memory and dedicated load/store DMA engines.
- Compiler: Wrote **custom MLIR/IREE passes** lowering ML ops onto the accelerator, plus a register/DMA interface allowing RTL datapath variants to be swapped without rebuilding the compiler.
- **Full-stack bring-up:** Built an end-to-end AI accelerator on a Xilinx Kria KV260 FPGA (Zynq SoC): custom compiler backend, C/C++ ARM-host runtime, and a double-buffered RTL shell with AXI DMA.

*LiteAIR5: Mixed-Precision AI Acceleration on a Custom **RISC-V** SoC* **IEEE SOCC 2023**

- Re-architected a 16-bit integer multiplier into a SIMD dot-product unit with selectable precisions, where lower precision yields proportionally higher throughput, integrated via RISC-V ISA and microarchitecture co-design.
- Integrated the extended core into a full FPGA-based System-on-Chip; delivered **15.5–47.7×** speedup and up to **20.5 TOPS/W** on quantized CNN kernels over the baseline RISC-V core.

EXPERIENCE

Research Assistant, High-Performance Low-Power (HPLP) Lab, UVA 2022–Present

- Hand-drew a balanced CMOS PUF macro in **GF12**; LVS/DRC-clean, targeting the Dec. 2026 shuttle.
- Teaching Assistant for Computer Architecture & Design, Embedded Systems, and AI Hardware courses.

SKILLS

RTL/Verification: SystemVerilog, Verilog, Vivado, ModelSim
Physical Design: Synopsys DC/ICC2, Cadence Virtuoso (ADE, Spectre), LVS/DRC signoff, GF12
Programming: C/C++, Python, RISC-V Assembly, CUDA, Tcl, Bash, Git, Make

SELECTED PUBLICATIONS & AWARDS

- **Y. Gao**, et al., "FlexPosit: Tunable Fractional Precision for LLM Inference Accelerators," **MICRO 2026**.
- J. Yin, L. Dai, **Y. Gao**, et al., "High-Performance Low-Power Adiabatic Systolic Array Design," **ICCD 2026 Best Paper Award**.
- X. Zhao, R. Xu, **Y. Gao**, et al., "Edge-MPQ: Layer-Wise Mixed-Precision Quantization," **IEEE TC**, 2024.
- **Y. Gao**, et al., "LiteAIR5: AI-Extended RISC-V Cores," **IEEE SoCC**, 2023.
- **2026 LAD Fellowship**: $10,000 award for LLM-driven NPU design space exploration (11 of 52 selected).
- **U.S. Patent Application**: Membrane DRAM-PIM Filtering Architecture, 2025 (Contributor).